
LAN86xx Bus Interface Network (BIN) Reference Design Application Note

INTRODUCTION

The purpose of this document is to communicate the latest design and PCB layout recommendations for the Bus Interface Network (BIN) used with the Microchip LAN86xx family of devices. The BIN is the network that interfaces the PHY IC to the MDI connector in 10BASE-T1S applications.

As with any design for automotive or industrial applications, it can be very challenging to achieve adequate *Signal Integrity* (SI) and *Electromagnetic Compatibility* (EMC) performance. Application boards should be designed with both signal integrity and EMC in mind. The PCB design must utilize proper grounding, power supply routing, decoupling, and a PCB layout strategy to manage EMC emissions and immunity.

OVERVIEW

This document includes the following chapters:

[Chapter 1.0: Application Circuit](#)

[Chapter 2.0: PCB Layout Guidelines](#)

[Chapter 3.0: PCB Layout Example](#)

[Appendix A: References](#)

[Appendix B: Application Note Revision History](#)

Each chapter provides design recommendations that must be carefully considered on a case-by-case basis depending on application requirements and limitations.

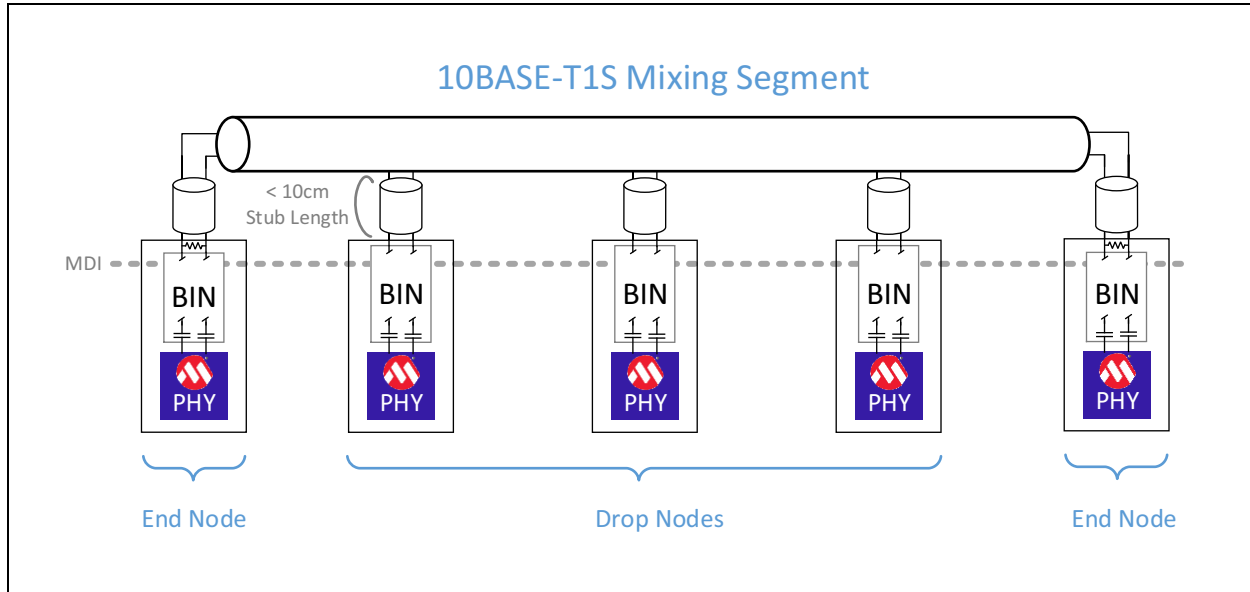
1.0 APPLICATION CIRCUIT

An example 10BASE-T1S single pair Ethernet mixing segment is shown below in [Figure 1-1](#). This network operates at 10 Mbit/s over a single pair of conductors, this is an end-terminated bus topology, with 100Ω nominal differential impedance. Nodes in the middle of the network are referred to as *Drop Nodes* while nodes at the end of the mixing segment are *End Nodes*. This document describes Microchip's recommendation for BIN implementation - the components and requirements between the MDI/Device connector and the PHY.

The BIN circuitry can be tailored to support a variety of industrial, automotive, and backplane Ethernet applications.

The component selections and rating will vary based upon the application as well as the OEM's EMC requirements, such as allowed levels for RF emissions and desired levels for immunity to RF stress.

FIGURE 1-1: NODE TOPOLOGY



1.1 MINIMAL BIN

The MINIMAL BIN application features an example of the minimum components required to pass basic industrial compliance testing. Application examples include industrial sensors/actuators, industrial control cabinets and machine control, and building automation.

FIGURE 1-2: MINIMAL BIN CIRCUITRY

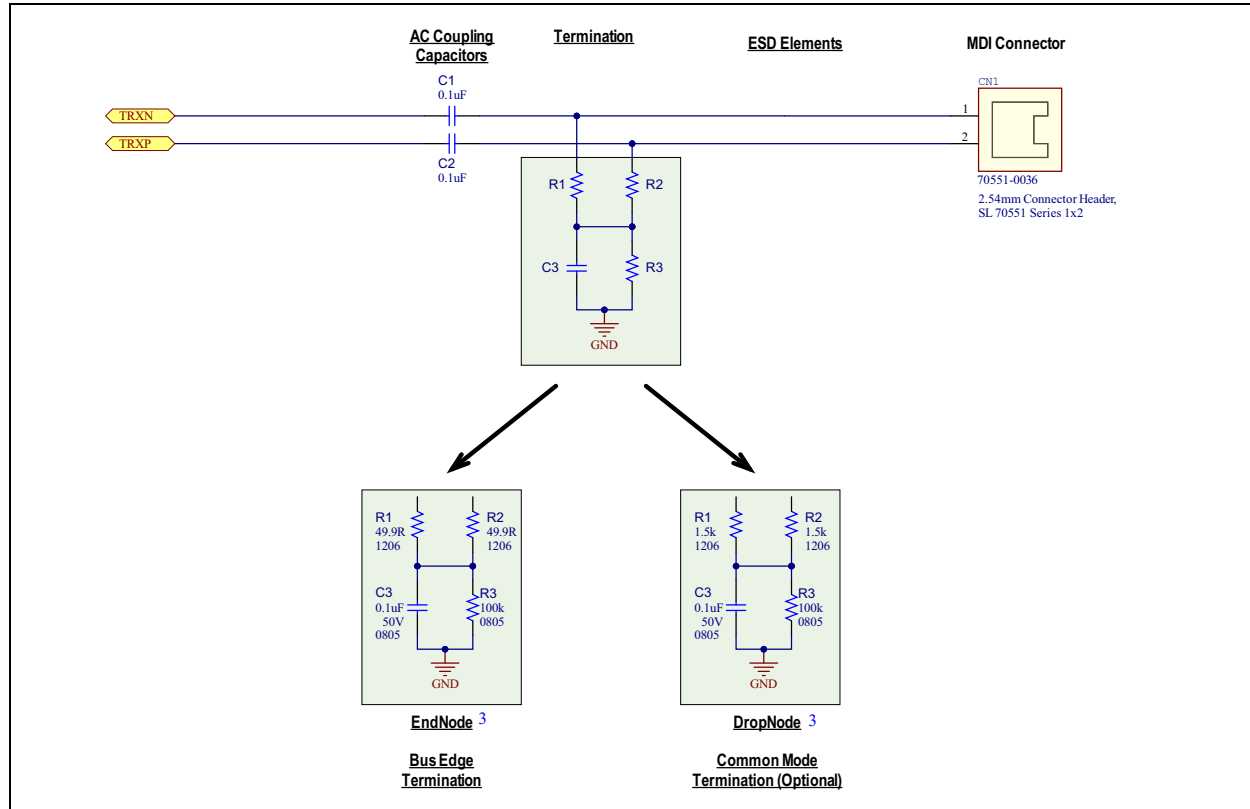


TABLE 1-1: MINIMAL BIN - COMPONENT SELECTION

Reference	Part Number ¹	Manufacturer	Description ²	Comments
C1,C2			CAP, 0.1 uF, 50V	
R1,R2			RES, 49.9Ω, 1%, 1206 (3216 metric)	End Node
			RES, 1.5 KΩ, 1206 (3216 metric)	Drop Node
C3			CAP, 0.1 uF, 50V, 0805 (2012 metric)	
R3			RES, 100 KΩ, 0805 (2012 metric)	
CN1			MDI Connector	

Note 1: Exact component part numbers and parameters must be determined by the customer's unique application requirements.

2: Components with a specific size listed, ex. 1206 (3216 metric) have been chosen specifically for EMC Susceptibility mitigation. Please do not deviate from these sizes.

3: Bus edge termination is required on End Nodes. Common mode termination is optional on Drop Nodes.

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AC Coupling Capacitors:

Components C1 and C2 are the AC coupling capacitors. These capacitors create Galvanic isolation between the MDI connector and the PHY and should be sized according to the application's EMI/EMC and system requirements. These capacitors must be installed on all nodes.

Termination:

Components R1, R2, R3, C3 terminate the bus, either for a *Drop Node* or an *End Node*. The termination on a *Drop Node* helps dampen common mode system resonances. The termination on an *End Node* performs signal integrity bus termination. The value of R1 and R2 differ between a *Drop Node* and an *End Node*.

Note: These resistors will also affect the differential impedance.

MDI Connector:

Component CN1 is the connector used to attach to the bus. There is no standard connector defined so the exact component part number must be determined by the customer's unique application requirements.

1.2 OPTIMIZED BIN

The OPTIMIZED BIN provides additional robustness and EMC performance that would be required for more stringent industrial or automotive applications.

FIGURE 1-3: OPTIMIZED BIN CIRCUITRY

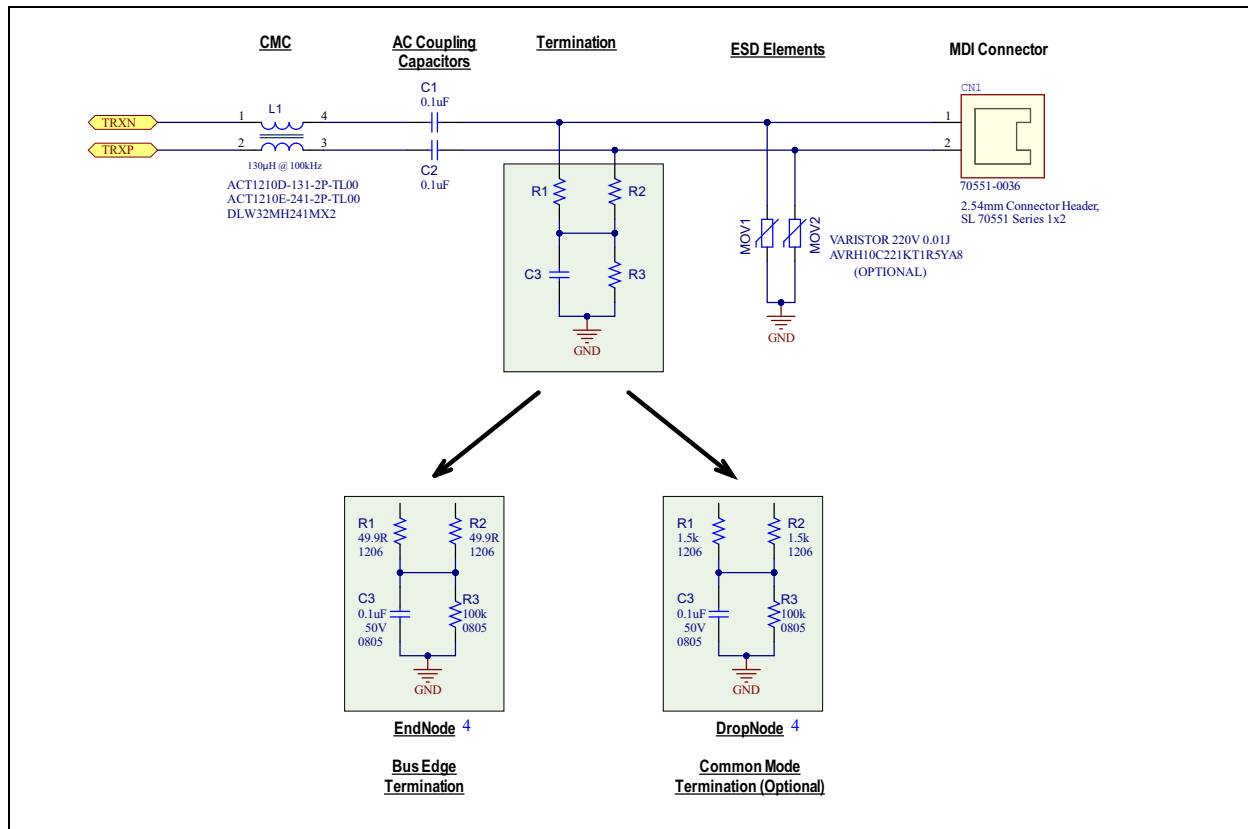


TABLE 1-2: OPTIMIZED BIN - COMPONENT SELECTION

Reference	Part Number ¹	Manufacturer	Description ^{2, 3}	Comments
C1,C2			CAP, 0.1 µF, 50V	
R1,R2			RES, 49.9Ω, 1%, 1206 (3216 metric)	End Node
			RES, 1.5 KΩ, 1206 (3216 metric)	Drop Node
R3			RES, 100 KΩ, 0805 (2012 metric)	
C3			CAP, 0.1 µF, 50V, 0805 (2012 metric)	
CN1			MDI Connector	
L1	ACT1210D-131-2P-TL00	TDK	CMC, 130 µH @ 100 kHz	
	ACT1210E-241-2P-TL00	TDK	CMC, 240 µH @ 100 kHz	
	DLW32MH241MX2	Murata	CMC, 240 µH @ 100 kHz	
MOV1, MOV2	AVRH10C221KT1R5YA8	TDK	Varistor, ESD Suppressor, 25 kV	
	EZA-EG3W11AV	Panasonic	TVS Diode, ESD Suppressor, 25 kV	

Note 1: Exact component part numbers and parameters must be determined by the customer's unique application requirements.

2: When multiple devices are listed in the description, all of them are acceptable for use.

3: Components with a specific size listed, ex. 1206 (3216 metric) have been chosen specifically for EMC Susceptibility mitigation. Please do not deviate from these sizes.

4: Bus edge termination is required on End Nodes. Common mode termination is optional on Drop Nodes.

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Common Mode Choke (CMC):

The CMC will pass on differential and suppress common mode signal components, resulting in increased EMC immunity and lower EM (electro-magnetic) emission levels. The exact component ratings and sizes are determined by the application and EMI/EMC and system requirements.

AC Coupling Capacitors:

Components C1 and C2 are the AC coupling capacitors. These capacitors create Galvanic isolation between the MDI connector and the PHY and should be sized according to the application's EMI/EMC and system requirements. These capacitors must be installed on all nodes.

Termination:

Components R1, R2, R3, C3 terminate the bus, either for a *Drop Node* or an *End Node*. The termination on a *Drop Node* helps dampen common mode system resonances. The termination on an *End Node* performs signal integrity bus termination. The value of R1 and R2 differ between a *Drop Node* and an *End Node*.

Note: These resistors will also affect the differential impedance.

ESD Elements:

Components MOV1 and MOV2 are for ESD protection and should be located near the MDI connector. The exact component ratings and sizes are determined by the application's EMI/EMC and system requirements. When selecting ESD protection, the diode's capacitive loading effects on the bus must be considered.

MDI Connector:

Component CN1 is the connector used to attach to the bus. There is no standard connector defined so the exact component part number must be determined by the customer's unique application requirements.

1.3 DEMO BIN

The DEMO BIN is used on some Microchip demo and evaluation boards when PoDL (Power over Data Line) is required. The use of PoDL requires a BIN that does not DC load the network. It is recommended that unless PoDL is required, the MINIMAL BIN or OPTIMIZED BIN should be used. Note at the time of this writing, there is no official standard for PoDL on 10BASE-T1S networks; it is considered a demo feature only.

FIGURE 1-4: DEMO BIN CIRCUITRY

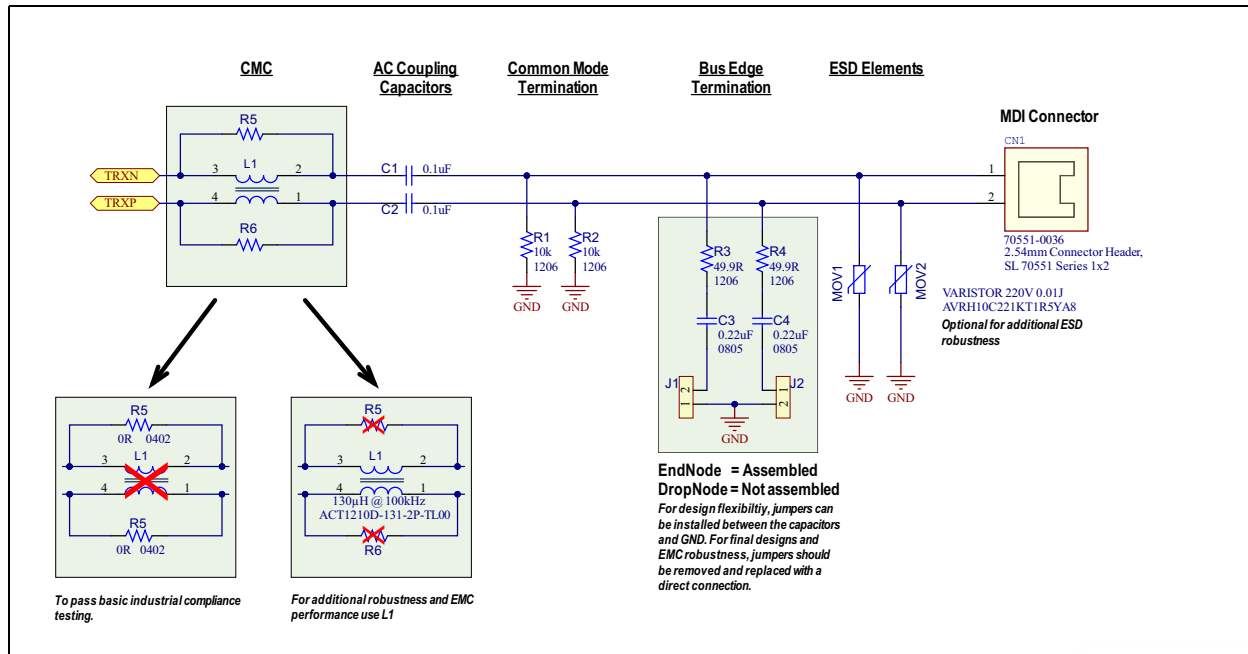


TABLE 1-3: DEMO BIN - COMPONENT SELECTION

Reference	Part Number ¹	Manufacturer	Description ^{2, 3}	Comments
C1,C2			CAP, 0.1 uF, 50V	
R1,R2			RES, 10 KΩ, 1206 (3216 metric)	
R3,R4			RES, 49.9Ω, 1%, 1206 (3216 metric)	Only installed on End Node
C3,C4			CAP, 0.22 uF, 50V, 0805 (2012 metric)	Only installed on End Node
L1	ACT1210D-131-2P-TL00	TDK	CMC, 130 uH @ 100 kHz	
R5,R6			RES, 0Ω, 0402 (1005 metric)	
MOV1, MOV2	AVRH10C221KT1R5YA8	TDK	Varistor, ESD Suppressor, 25 kV	
	EZA-EG3W11AV	Panasonic	TVS Diode, ESD Suppressor, 25 kV	

Note 1: Exact component part numbers and parameters must be determined by the customer's unique application requirements.

2: When multiple devices are listed in the description, all of them are acceptable for use.

3: Components with a specific size listed, ex. 1206 (3216 metric) have been chosen specifically for EMC Susceptibility mitigation. Please do not deviate from these sizes.

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Common Mode Choke (CMC):

The Common Mode Choke (CMC) will pass on differential and suppress common mode signal components, resulting in increased EMC immunity and lower EM (electro-magnetic) emission levels. The exact component ratings and sizes are determined by the application and EMI/EMC and system requirements.

AC Coupling Capacitors:

Components C1 and C2 are the AC coupling capacitors. These capacitors create Galvanic isolation between the MDI connector and the PHY and should be sized according to the application's EMI/EMC and system requirements. These capacitors must be installed on all nodes.

Common Mode Termination:

Components R1, R2 are installed on both *Drop Nodes* and *End Nodes* to dampen common mode system resonances.

Bus Edge Termination:

Components R3, R4, C3, C4 are installed on *End Nodes* only to perform signal integrity bus termination.

ESD Elements:

Components MOV1 and MOV2 are for ESD protection and should be located near the MDI connector. The exact component ratings and sizes are determined by the application's EMI/EMC and system requirements. When selecting ESD protection, the diode's capacitive loading effects on the bus must be considered.

MDI Connector:

Component CN1 is the connector used to attach to the bus. There is no standard connector defined so the exact component part number must be determined by the customer's unique application requirements.

2.0 PCB LAYOUT GUIDELINES

Proper PCB hardware implementation (e.g. component placement, signal routing and shield/chassis/PCB grounding) is essential for achieving optimal EMC performance. This section provides PCB layout guidelines for LAN86xx BIN applications.

1. Per [IEEE Std 802.3cg™-2019, IEEE Standard for Ethernet, Amendment 5 \[1\]](#) the stub length between the mixing segment and the MDI/connector should not exceed 10 cm.
2. Board traces between the LAN86xx pins (TRXP/TRXN) and the MDI connector (CN1) must be implemented as length matched, impedance controlled, 50Ω single-ended traces (rather than 100Ω differential traces) to minimize differential capacitance to 15 pF per Clause 147.9.2, Table 147-4 of [IEEE Std 802.3cg™-2019, IEEE Standard for Ethernet, Amendment 5 \[1\]](#). Note that keeping capacitance low is more critical than maintaining exact impedance. In [Section 3.0, PCB Layout Example](#), impedance is roughly maintained via coplanar techniques rather than a typical reference layer approach.
3. All traces between the LAN86xx pins and the MDI connector must be kept as short as possible with sufficient clearance (at least three times the trace width, measured center to center,) to other signals or metallic features, to mitigate crosstalk interference.
4. All BIN components between the LAN86xx pins and the MDI connector should be placed closely together with minimized trace lengths to reduce capacitance.
 - a. CMC (L1) and AC coupling capacitors (C1 and C2) must be placed as close as possible to the MDI connector.
 - b. ESD and termination elements should be placed in-line, so that stubs are minimized. ESD elements should be placed as close as possible to their respective MDI connector pin.
5. Care should be taken regarding potential ground flooding on the outside layer surrounding the BIN as this can increase the trace capacitance. If ground flooding is used, a void in the ground flooding on the outside layer surrounding the BIN should be implemented. An example of such a void is shown in [Section 3.0, PCB Layout Example](#).
6. An all-layer void underneath the CMC (L1) should be used to prevent any coupling from the CMC core and coils to/from the internal planes and traces.

Disclaimer:

- Components and values will be updated as BIN validation and EMC testing continues.
- The customer is responsible for determining suitability of components for their application purpose.
- The customer must certify and ensure the design system functionality conformance.

3.0 PCB LAYOUT EXAMPLE

This section contains a PCB layout example implementation from the [Microchip LAN8670 USB Evaluation Board \[2\]](#). This Evaluation Board implements the DEMO BIN from [Section 1.3, DEMO BIN](#). This board also follows the guidelines in [Section 2.0, PCB Layout Guidelines](#). The PCB stack-up is a 4-layer arrangement, a description of each layer is provided in [Figure 3-1](#) and [Table 3-1](#). Please also see the [LAN8670/1/2 10BASE-T1S Ethernet PHY Transceiver Data Sheet \[3\]](#).

FIGURE 3-1: EVB08L38 LAN8670 10BASE-T1S-TO-USB BOARD STACKUP

Board Stack Report					
Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
		Top Paste			
		Top Overlay			
		Top Solder	Solder Resist		3.5
1		Top Layer	Copper	0.018mm	
		Dielectric 1	2x1080	0.140mm	4.2
2		GND	Copper	0.033mm	
		Dielectric 2	FR-4	0.930mm	4.2
3		Power	Copper	0.033mm	
		Dielectric 3	2x1080	0.140mm	4.2
4		Bottom Layer	Copper	0.018mm	
		Bottom Solder	Solder Resist		3.5
		Bottom Overlay			
		Bottom Paste			

Height : 1.312mm

TABLE 3-1: PCB LAYER DESCRIPTION

Layer	Layer Name	Description	Figure
1	Layer 1 (Top)	Top side components and signal routing. This layer also has PCB ground flooding.	Figure 3-2
2	Layer 2	Inner plane layer used as a ground plane.	Figure 3-3
3	Layer 3	Inner plane layer used for power and ground.	Figure 3-4
4	Layer 4 (Bottom)	Bottom side components and signal routing. This layer also has PCB ground flooding.	Figure 3-5

3.1 Example LAN8670 USB Evaluation Board EVB-LAN8670-USB (EV08L38A)

[Figure 3-2](#) through [Figure 3-5](#) show the layers from the Microchip LAN8670 USB Evaluation Board, EV08L38A.

FIGURE 3-2: EV08L38A LAYER 1

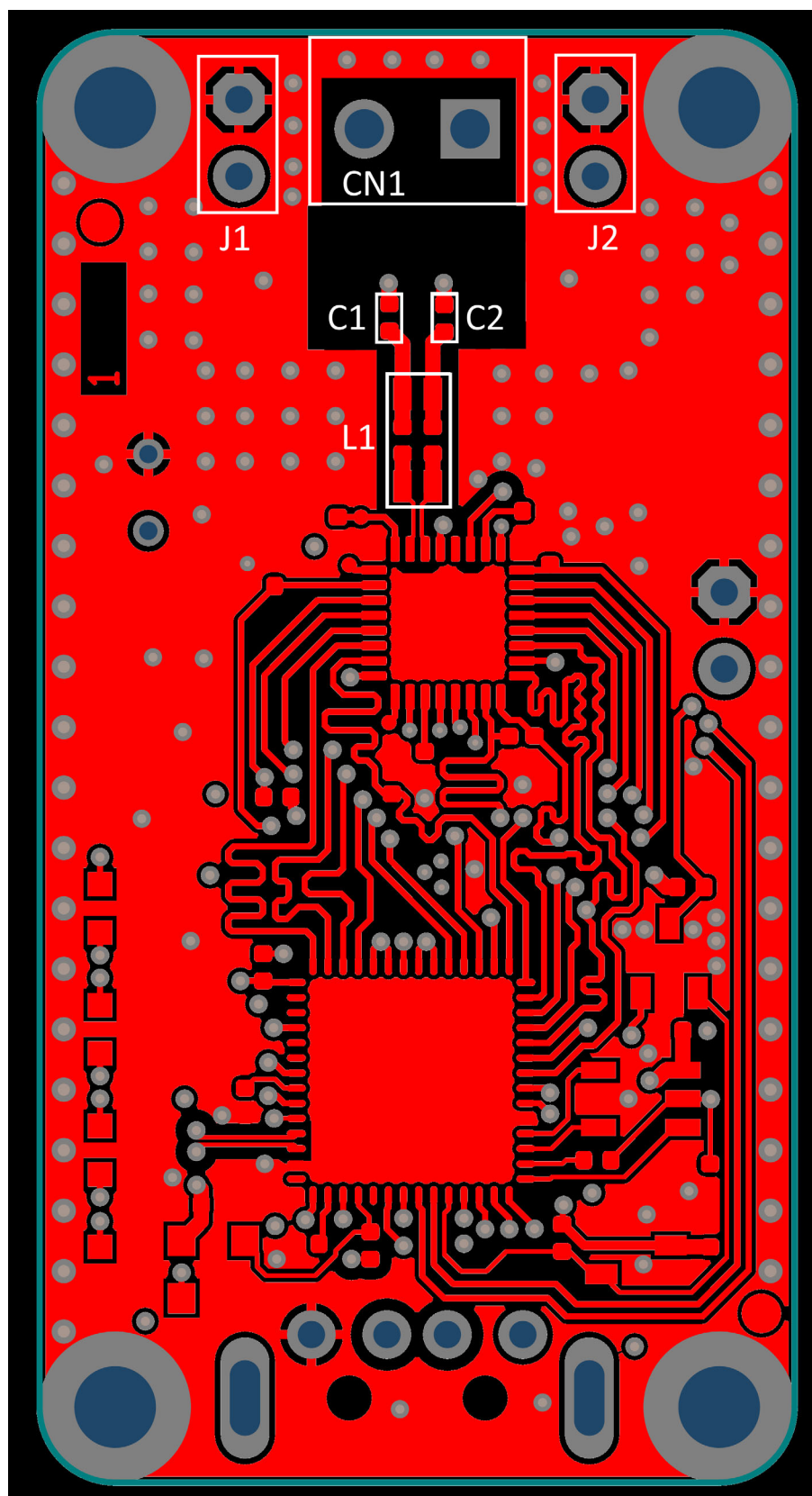


FIGURE 3-3: EV08L38A LAYER 2

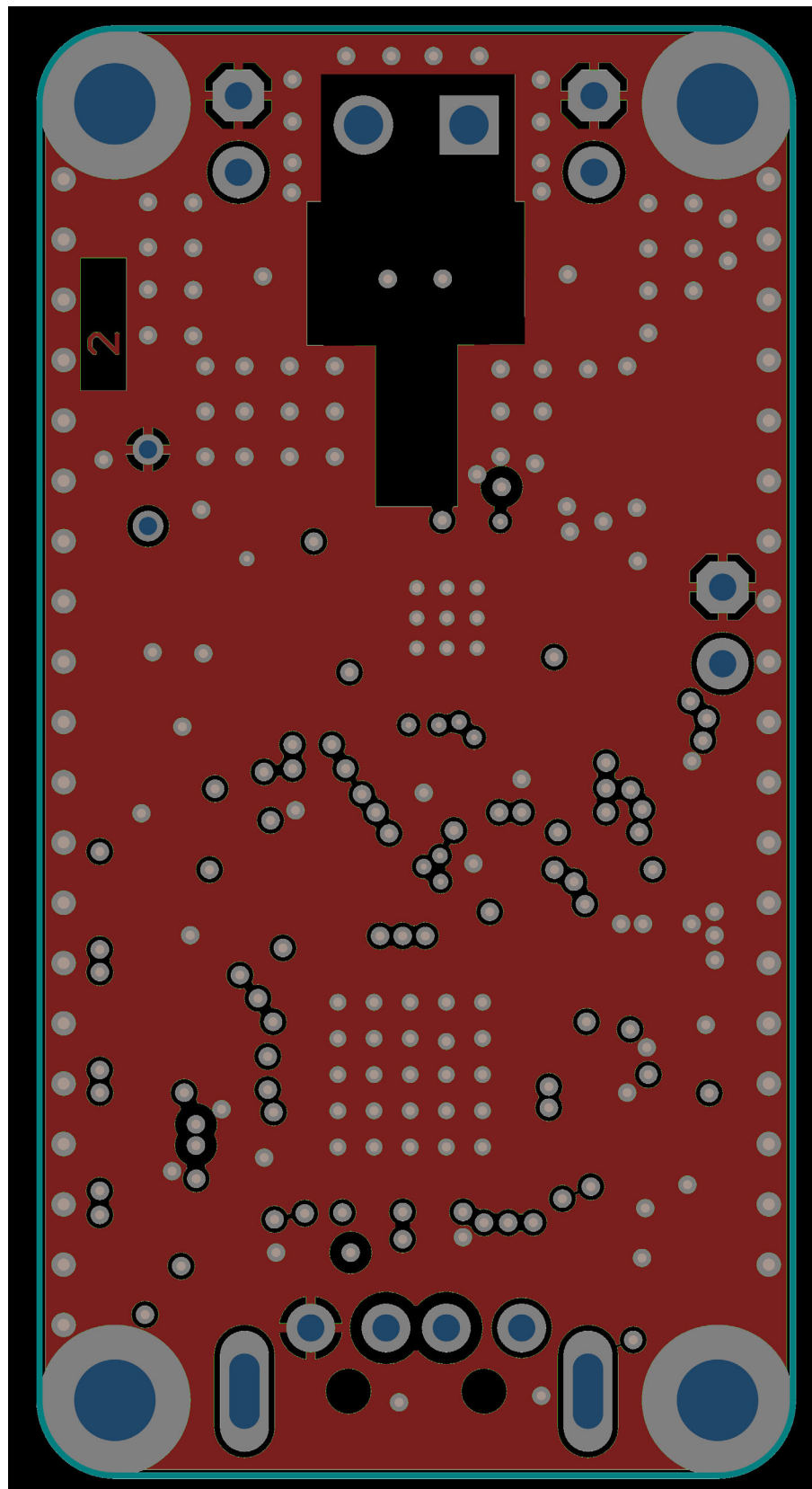


FIGURE 3-4: EV08L38A LAYER 3

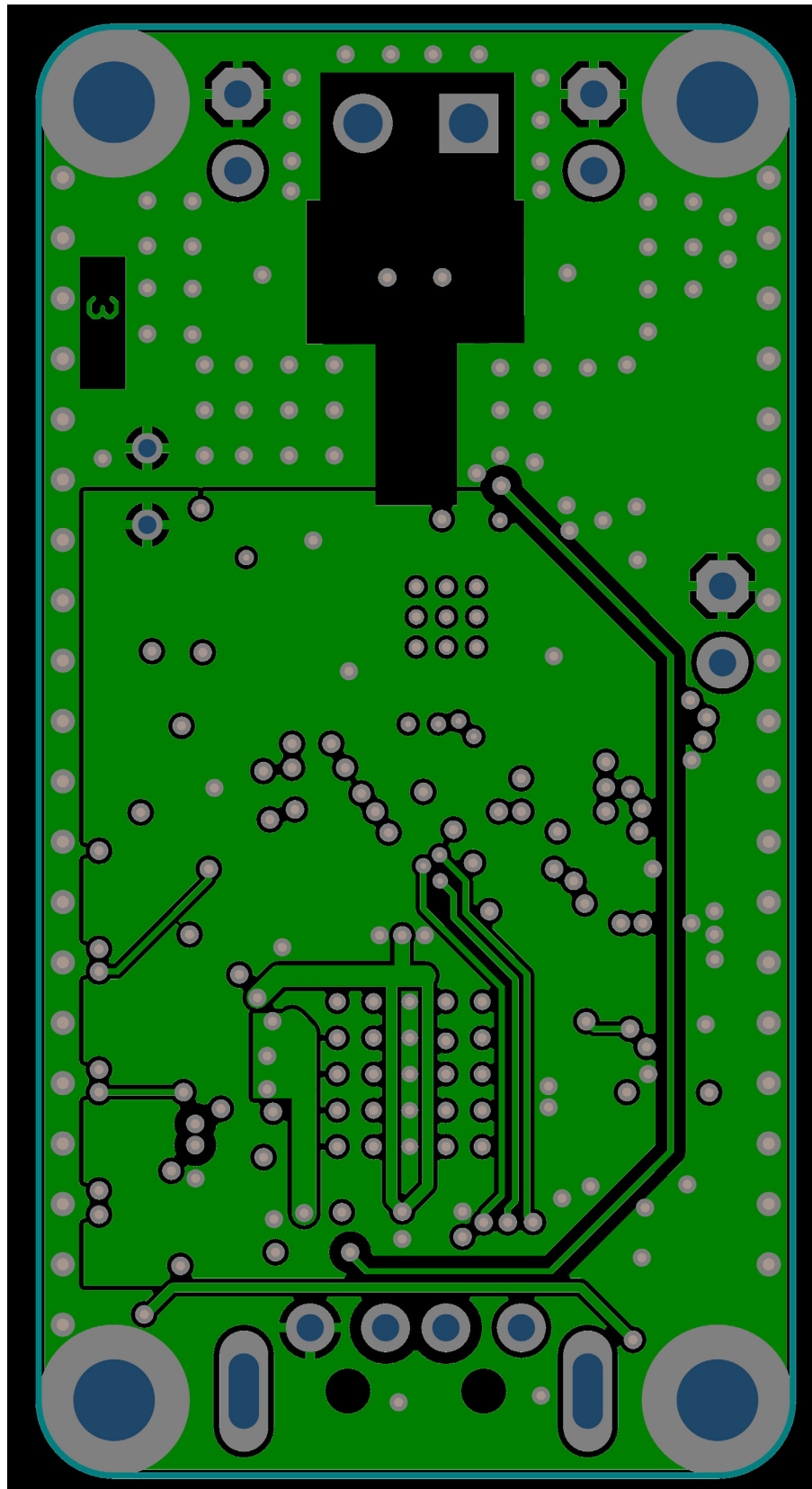
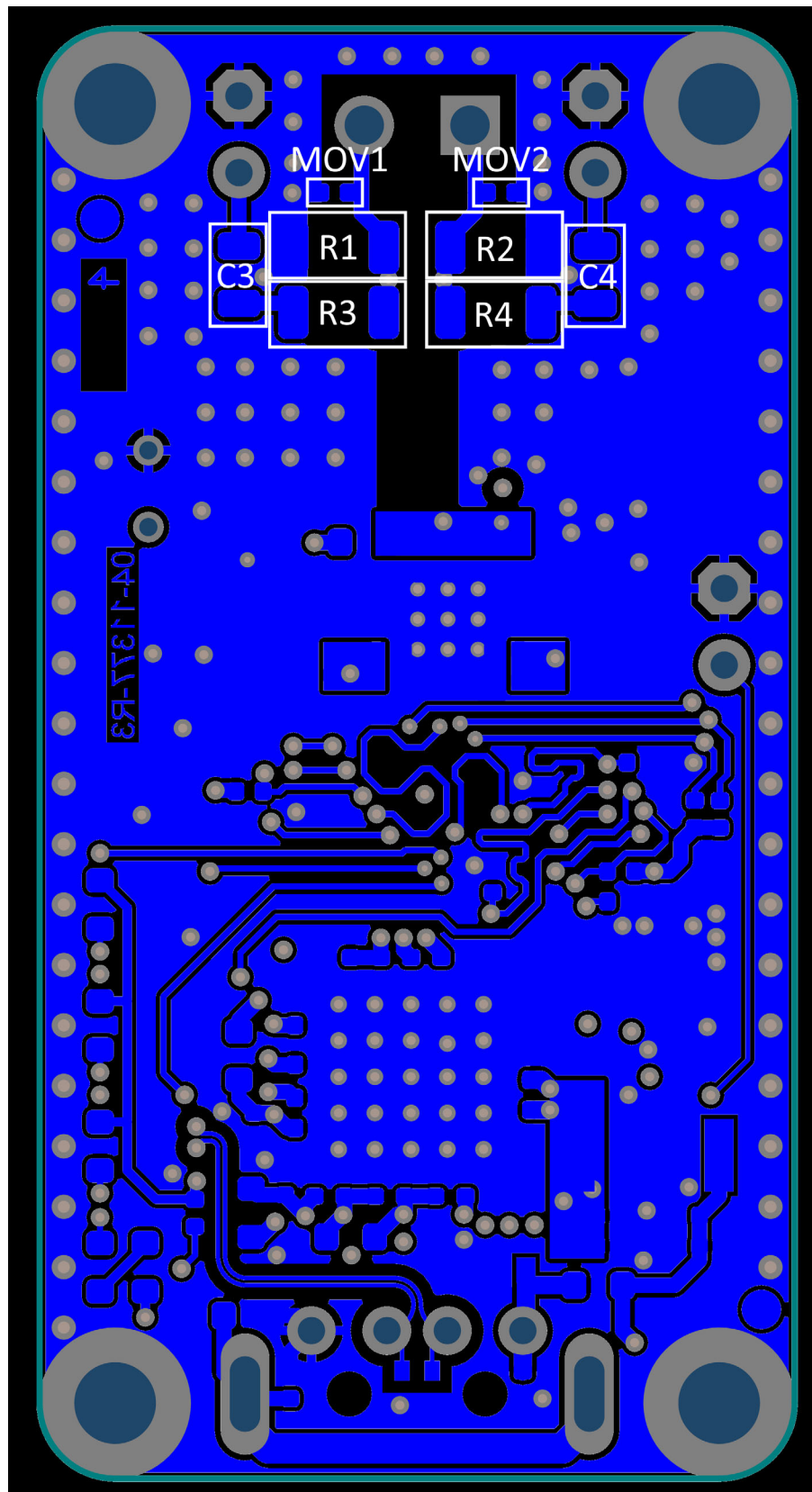


FIGURE 3-5: EV08L38A LAYER 4



APPENDIX A: REFERENCES

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- [1] IEEE Std 802.3cg™-2019, IEEE Standard for Ethernet, Amendment 5
Physical Layer Specifications and Management Parameters for 10 Mb/s Operation and Associated Power Delivery over a Single Balanced Pair of Conductors.
standards.ieee.org/standard/802_3cg-2019.html
- [2] Microchip LAN8670 USB Evaluation Board
EVB-LAN8670-USB (EV08L38A). Microchip. www.microchip.com.
- [3] LAN8670/1/2 10BASE-T1S Ethernet PHY Transceiver Data Sheet
60001573. Microchip. www.microchip.com.

APPENDIX B: APPLICATION NOTE REVISION HISTORY

B.1 Current Document Revision

Revision C (DS60001718C - April 2024)

- Updated definitions and schematics of OPTIMIZED BIN

B.2 Previous Document Revision

Revision B (DS60001718B - October 2022)

- Updated definitions and schematics of MINIMAL BIN and OPTIMIZED BIN
- Added DEMO BIN
- Removed BACKPLANE BIN

Revision A (DS60001718A - September 2021)

- Initial Release

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